

第二章习题

Lemma: 设偏序集 $(P, \leq)$ 的所有非空子集皆有极大元. 若单射 $\theta: P \rightarrow P$ 满足 $\forall x \in P, x \leq \theta(x)$, 则有: $\theta = id_P$

Proof: 令 $S = \{x \in P : \theta(x) \neq x\}$. $\therefore S \subseteq P$.

假设 $S \neq \emptyset$, 则有: S 是 P 的非空子集. $\therefore S$ 有极大元.

设 λ 为 S 的极大元. $\therefore \lambda \in S$, 且不存在 $\gamma \in S$ 使得 $\gamma > \lambda$.

$\therefore \lambda \in S \subseteq P$ $\therefore \lambda \in P$ $\therefore \lambda \leq \theta(\lambda)$

$\therefore \lambda \in S$ $\therefore \theta(\lambda) \neq \lambda$ $\therefore \lambda \neq \theta(\lambda)$ $\therefore \lambda < \theta(\lambda)$ $\therefore \theta(\lambda) > \lambda$

$\therefore \lambda \in P$, $\theta: P \rightarrow P$ 是映射 $\therefore \theta(\lambda) \in P$. $\therefore \theta(\theta(\lambda)) \in P$.

假设 $\theta(\theta(\lambda)) = \theta(\lambda)$, 则 $\therefore \theta: P \rightarrow P$ 是单射 $\therefore \theta(\lambda) = \lambda$

$\therefore \lambda = \theta(\lambda)$ 与 $\lambda \neq \theta(\lambda)$ 矛盾.

$\therefore \theta(\theta(\lambda)) \neq \theta(\lambda)$ $\therefore \theta(\lambda) \in S$.

$\therefore \theta(\lambda) \in S$ 且 $\theta(\lambda) > \lambda$ 矛盾. $\therefore S = \emptyset$.

$\therefore \forall x \in P$, 有: $\theta(x) \in P$, 且 $\theta(x) = x$.

$\therefore \theta = id_P$. $\square$

习题: 设 $(A, \leq)$ 为预序集. 定义 A 上的二元关系 $\sim$ 如下:

$\forall a, a' \in A$, 有: $a \sim a' \iff (a \leq a') \wedge (a' \leq a)$

(i) 说明 $\sim$ 是集合 A 上的等价关系.

Proof: $\forall a \in A$, 有: $\therefore (A, \leq)$ 是预序集 $\therefore a \leq a$ $\therefore (a \leq a) \wedge (a \leq a)$

$\therefore a \sim a$ $\therefore$ 反身性成立.

对 $\forall a, b \in A$. 若 $a \sim b$, 则有:

$$\therefore a \sim b \quad \therefore (a \leq b) \wedge (b \leq a) \quad \therefore (b \leq a) \wedge (a \leq b) \quad \therefore b \sim a$$

$\therefore$ 对称性成立.

对 $\forall a, b, c \in A$. 若 $a \sim b$ 且 $b \sim c$, 则有:

$$\therefore a \sim b \quad \therefore (a \leq b) \wedge (b \leq a)$$

$$\therefore b \sim c \quad \therefore (b \leq c) \wedge (c \leq b)$$

$$\therefore a \leq b \text{ 且 } b \leq c \quad \therefore a \leq c$$

$$\therefore c \leq b \text{ 且 } b \leq a \quad \therefore c \leq a$$

$$\therefore (a \leq c) \wedge (c \leq a) \quad \therefore a \sim c \quad \therefore \text{传递性成立.}$$

$\therefore \sim$ 是集合 A 上的等价关系. $\square$

Remark: 对 $\forall a \in A$, 记包含 a 的等价类为 $[a]$. 则有: $A/\sim = \{[a] \mid a \in A\}$

$$\text{对 } \forall a \in A, [a] = \{x \in A : x \sim a\} = \{x \in A : (x \leq a) \wedge (a \leq x)\}$$

在 $A/\sim$ 上定义序结构为: 对 $\forall [a], [a'] \in A/\sim$ (其中 $a, a' \in A$),
 $[a] \leq [a'] \iff a \leq a'$

Lemma: 对 $\forall [a], [a'] \in A/\sim$ (其中 $a, a' \in A$), 有:

$$[a] \leq [a'] \iff \text{对 } \forall x \in [a], \forall y \in [a'], \text{ 有: } x \leq y$$

$$\text{Proof: } (\Leftarrow): \therefore a \in [a], a' \in [a'] \quad \therefore a \leq a' \quad \therefore [a] \leq [a']$$

$(\Rightarrow):$ 对 $\forall x \in [a], \forall y \in [a']$, 有:

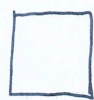
$$\therefore x \in [a] \quad \therefore x \in A \text{ 且 } x \sim a \quad \therefore x \leq a \text{ 且 } a \leq x$$

$$\therefore y \in [a'] \quad \therefore y \in A \text{ 且 } y \sim a' \quad \therefore y \leq a' \text{ 且 } a' \leq y$$

$$\because [a] \leq [a'] \quad \therefore a \leq a'$$

$$\because x \leq a \text{ 且 } a \leq a' \quad \therefore x \leq a'$$

$$\because x \leq a' \text{ 且 } a' \leq y \quad \therefore x \leq y$$



Lemma: $(A/\sim, \leq)$ 是偏序集.

Proof: 对 $\forall [a] \in A/\sim$ (其中 $a \in A$), $\because a \leq a \quad \therefore [a] \leq [a]$.

$\therefore$ 反身性成立.

对 $\forall [a], [b], [c] \in A/\sim$ (其中 $a, b, c \in A$), 若 $[a] \leq [b]$ 且 $[b] \leq [c]$,

则有: $\because [a] \leq [b] \quad \therefore a \leq b$

$\because [b] \leq [c] \quad \therefore b \leq c$

$\because a \leq b \text{ 且 } b \leq c \quad \therefore a \leq c \quad \therefore [a] \leq [c] \quad \therefore$ 传递性成立.

对 $\forall [a], [b] \in A/\sim$ (其中 $a, b \in A$), 若 $[a] \leq [b]$ 且 $[b] \leq [a]$, 则有:

$\because [a] \leq [b] \quad \therefore a \leq b \quad \because [b] \leq [a] \quad \therefore b \leq a$

$\because a \leq b \text{ 且 } b \leq a \quad \therefore a \sim b \quad \therefore [a] = [b] \quad \therefore$ 反称性成立.

$\therefore (A/\sim, \leq)$ 是偏序集

Lemma: 商映射 $q: A \rightarrow A/\sim$ 是严格保序的映射.

Proof: $(A, \leq)$ 是预序集, $(A/\sim, \leq)$ 是偏序集, 商映射

$q: A \rightarrow A/\sim$ 是预序集之间的映射.

对 $\forall a, a' \in A$.

若 $a \leq a'$, 则有: $[a] \leq [a'] \quad \therefore q(a) = [a] \leq [a'] = q(a')$

若 $q(a) \leq q(a')$, 则有: $[a] \leq [a'] \quad \therefore a \leq a'$

$\therefore a \leq a' \iff q(a) \leq q(a') \quad \therefore q: A \rightarrow A/\sim$ 是严格保序的映射. □

(iii) 对任意偏序集 $(B, \leq)$, 明确地写下一一对应

$$\{f: A \rightarrow B \mid \text{保序映射}\} \xleftrightarrow{1:1} \{\bar{f}: (A/\sim) \rightarrow B \mid \text{保序映射}\}$$

Proof: 定义映射 $\varphi: \{f: A \rightarrow B \mid \text{保序映射}\} \longrightarrow \{\bar{f}: (A/\sim) \rightarrow B \mid \text{保序映射}\}$ 如下:

对 $\forall f: A \rightarrow B$ 是保序映射.

$(A, \leq)$ 是预序集, $\sim$ 是集合 A 上的等价关系, $q: A \rightarrow A/\sim$ 是对应的商映射.

$(B, \leq)$ 是任意的偏序集, $f: A \rightarrow B$ 是一个映射, 且满足对 $\forall a, a' \in A$,

若 $a \sim a'$, 则有: $\therefore a \sim a' \quad \therefore a \leq a' \text{ 且 } a' \leq a$

$\therefore a \leq a', f: A \rightarrow B$ 是保序映射 $\therefore f(a) \leq f(a')$

$\therefore a' \leq a, f: A \rightarrow B$ 是保序映射 $\therefore f(a') \leq f(a)$

$\therefore f(a) \leq f(a') \text{ 且 } f(a') \leq f(a), (B, \leq)$ 是偏序集 $\therefore f(a) = f(a')$

$\therefore$ 存在唯一的映射 $\bar{f}: (A/\sim) \rightarrow B$ 使得 $\bar{f} \circ q = f$.

对 $\forall [a], [a'] \in A/\sim$ (其中 $a, a' \in A$). 若 $[a] \leq [a']$, 则有: $a \leq a'$

$\therefore f(a) \leq f(a')$

$\therefore \bar{f}([a]) = \bar{f}(q(a)) = (\bar{f} \circ q)(a) = f(a) \leq f(a') = (\bar{f} \circ q)(a') = \bar{f}(q(a')) = \bar{f}([a'])$

$\therefore \bar{f}: (A/\sim) \rightarrow B$ 是保序映射.

令 $\varphi(f) = \bar{f} \in \{\bar{f}: (A/\sim) \rightarrow B \mid \text{保序映射}\}$

对 $\forall f_1, f_2 \in \{f: A \rightarrow B \mid \text{保序映射}\}$.

若 $f_1 = f_2$, 则有: $\therefore f_1: A \rightarrow B$ 是保序映射 $\therefore \varphi(f_1) = \bar{f}_1: (A/\sim) \rightarrow B$ 是保序映射

且有 $\bar{f}_1 \circ q = f_1$

$\therefore f_2: A \rightarrow B$ 是保序映射 $\therefore \varphi(f_2) = \bar{f}_2: (A/\sim) \rightarrow B$ 是保序映射,

且有 $\bar{f}_2 \circ q = f_2$

对 $\forall [a] \in A/\sim$ (其中 $a \in A$), 有: $\bar{f}_1([a]) = \bar{f}_1(q(a)) = (\bar{f}_1 \circ q)(a)$
 $= f_1(a) = f_2(a) = (\bar{f}_2 \circ q)(a) = \bar{f}_2(q(a)) = \bar{f}_2([a])$

$\therefore \bar{f}_1 = \bar{f}_2 \quad \therefore \varphi(f_1) = \varphi(f_2) \quad \therefore \varphi$ 是映射.

若 $\varphi(f_1) = \varphi(f_2)$, 则有: $\bar{f}_1 = \bar{f}_2 \quad \therefore f_1 = \bar{f}_1 \circ q = \bar{f}_2 \circ q = f_2$

$\therefore \varphi$ 是单射

对 $\forall \bar{f} \in \{\bar{f}: (A/\sim) \rightarrow B \mid \text{保序映射}\}$, 有: $\bar{f}: (A/\sim) \rightarrow B$ 是一个保序映射.

$\therefore q: A \rightarrow A/\sim$ 是对应的商映射. $\therefore \bar{f} \circ q: A \rightarrow B$ 是一个映射.

令 $f = \bar{f} \circ q \quad \therefore f: A \rightarrow B$ 是一个映射.

$\therefore (A, \leq)$ 是预序集, $(B, \leq)$ 是偏序集 $\therefore f: A \rightarrow B$ 是预序集之间的映射.

对 $\forall a, a' \in A$, 若 $a \leq a'$, 则有: $[a] \leq [a']$.

$\therefore \bar{f}: (A/\sim) \rightarrow B$ 是一个保序映射, $[a] \in A/\sim$, $[a'] \in A/\sim$,

$$[a] \leq [a']$$

$$\therefore \bar{f}([a]) \leq \bar{f}([a'])$$

$$\therefore f(a) = (\bar{f} \circ q)(a) = \bar{f}(q(a)) = \bar{f}([a]) \leq \bar{f}([a']) = \bar{f}(q(a')) = (\bar{f} \circ q)(a') = f(a')$$

$\therefore f: A \rightarrow B$ 是一个保序映射. $\therefore f \in \{f: A \rightarrow B \mid \text{保序映射}\}$

$(A, \leq)$ 是预序集, $\sim$ 是集合 A 上的等价关系, $q: A \rightarrow A/\sim$ 是对应的商映射.

$(B, \leq)$ 是任意的偏序集, $f: A \rightarrow B$ 是一个映射, 且满足: 对 $\forall a, a' \in A$,

若 $a \sim a'$, 则有: $\therefore a \sim a' \quad \therefore a \leq a' \text{ 且 } a' \leq a$

$\therefore a \leq a'$, $f: A \rightarrow B$ 是一个保序映射 $\therefore f(a) \leq f(a')$

$\therefore a' \leq a$, $f: A \rightarrow B$ 是一个保序映射 $\therefore f(a') \leq f(a)$

$\therefore f(a) \leq f(a')$ 且 $f(a') \leq f(a)$, $(B, \leq)$ 是偏序集 $\therefore f(a) = f(a')$

$\therefore$ 存在唯一的映射 $\bar{\eta}: (A/\sim) \rightarrow B$ 使得 $\bar{\eta} \circ q = f$

$\therefore \bar{f} \circ q = f, \bar{\eta} \circ q = f \quad \therefore \bar{f} = \bar{\eta} \quad \therefore \bar{\eta}: (A/\sim) \rightarrow B$ 是一个保序映射

$\therefore \varphi(f) = \bar{\eta} = \bar{f} \quad \therefore \varphi$ 是满射.

$\therefore \varphi$ 是双射.

